

FastAPI Framework Documentation

Official Reference Guide to High-Performance Asynchronous APIs and Pydantic

1. FastAPI Architecture & Routing

FastAPI is a modern, high-performance web framework for building REST APIs with Python based on OpenAPI, JSON Schema, Pydantic, and Starlette. It utilizes standard `async/await` coroutines and Uvicorn ASGI server.

```
from fastapi import FastAPI, HTTPException, Depends
from pydantic import BaseModel, EmailStr

app = FastAPI(title="Production API", version="1.0.0")

class UserCreate(BaseModel):
    username: str
    email: EmailStr

@app.post("/users/", status_code=201)
async def create_user(user: UserCreate):
    if user.username == "admin":
        raise HTTPException(status_code=400, detail="Username reserved")
    return {"status": "success", "user": user.model_dump()}
```

2. Dependency Injection & Security

FastAPI provides a powerful Dependency Injection system for database connections, authentication, and request scope validation via the `Depends()` decorator.

```
async def get_db_session():
    db = "DatabaseSessionConnection"
    try:
        yield db
    finally:
        print("Closed DB session")

@app.get("/items/")
async def list_items(db: str = Depends(get_db_session)):
    return {"db": db, "items": ["Item 1", "Item 2"]}
```